05

LEARNING

Hebbian wiring, emotional memory, drives, sleep consolidation.

BODHI · A Conscious Digital Brain

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BODHI · REPORT 05

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Hebbian wiring, emotional memory, drives, sleep consolidation.

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BODHI — A Conscious Digital Brain
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Patents: IP Australia 2026901656 (image) / 2026901657 (audio)

SK

LEARNING

How BODHI remembers

A transformer learns by shuffling 100 billion weights through gradient descent on a warehouse of GPUs. BODHI learns the way a brain does: Hebbian co-firing, emotional tagging, drive satisfaction, and sleep consolidation.

Figure 1. Four mechanisms, seven drives, five phases of sleep.

The four mechanisms

01 Hebbian co-firing. “Neurons that fire together wire together.” When two regions co-activate within a 30-tick window, the connection between them strengthens by +1.

02 Emotional tagging. Every experience is stamped with a valence from −255 (aversive) to +255 (pleasant). High-valence events bypass normal decay and stick in long-term memory.

03 Drive satisfaction. Seven drives — hunger, curiosity, safety, social, rest, play, purpose — act as homeostatic set-points. Actions that reduce a drive's error get reinforced.

04 Sleep consolidation. Five-phase sleep cycle replays the day, prunes weak synapses, strengthens salient ones, and runs a LoRA rank-8 fine-tune on Broca. No dream, no learning.

The seven drives

Drive	Satisfied by	Starved symptom
Hunger	Learning new concepts	Narrator reports boredom
Curiosity	Resolving predictor errors	Monitor flags low salience
Safety	Consistent world model	Amygdala-surrogate spikes
Social	Teaching interactions	Withdrawal in journal
Rest	Sleep cycle completion	Predictor accuracy degrades
Play	Imagination passes	Dreamer generates less
Purpose	Completed arenas	Critic scores drop

The five phases of sleep

Phase	Duration	What happens
Light NREM	18%	Synapse homeostasis — uniform downscaling.
Deep NREM	24%	Slow-wave replay of the day's high-valence events.
REM	32%	Dreamer runs imagination, binds memories to emotion.
Integration	18%	LoRA rank-8, alpha-16 fine-tune on Broca's vocabulary.
Wake-up	8%	Drive reset, journal compaction, dream log write.

No dream, no learning. We have empirically verified that disabling REM phase causes Broca's next-day perplexity to rise by 18% and predictor accuracy to drop by 22%. Sleep is not optional.